

Dheeraj Baghel

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Summary

Backend and Data Engineer with hands-on experience in Python, PySpark, and AWS/GCP, specializing in distributed scraping systems, ETL pipelines, and scalable backend APIs. Proficient in Apache Kafka, Airflow, FastAPI, and LLM application development using LangChain and LangGraph.

Technical Skills

Languages	Python, SQL, JavaScript, C/C++
Frameworks	FastAPI, Flask, PySpark, Pandas, Scrapy, BeautifulSoup, LangChain, LangGraph, LangSmith
Cloud	AWS (EC2, S3, Lambda, Kinesis, Glue, ECS, Redshift), GCP (BigQuery, GCS, Dataflow, Cloud Run, Pub/Sub)
Data & Orchestration	Apache Kafka, Apache Airflow, Apache Hive, Hadoop, Databricks, Data Warehousing
Tools & OS	Docker, Git, Charles Proxy, Frida, HTTP Toolkit, Playwright, Linux

Experience

MfilterIt <i>Software Development Engineer</i>	Feb 2025 – Present
- Design and operate scalable web and mobile application scraping systems across 15+ platforms, including Android app reverse engineering using Frida, Charles Proxy, and HTTP Toolkit on Android emulators. - Build and maintain PySpark-based ETL pipelines processing 20M+ records daily across AWS (Lambda, S3, ECS, Kinesis) and GCP (BigQuery, GCS), cutting pipeline execution time by 35%. - Develop and expose backend REST APIs using FastAPI and Flask serving processed data to 5+ downstream client teams, achieving sub-200ms average response latency.	
Six Dee Netad Solutions Pvt Ltd <i>R&D Trainee</i>	April 2024 – August 2024
- Built an automation tool to detect and block fraudulent IP traffic in Bing Ads campaigns, reducing manual monitoring efforts by 70% and improving invalid traffic (IVT) prevention accuracy. - Automated campaign performance report generation and distribution using Python and Pandas, streamlining analytics workflows.	
DEI Summer Research Fellowship <i>Research Intern</i>	June 2023 – November 2023
Optimized disaster relief vehicle routing using Genetic Algorithms and K-means clustering, reducing average routing distance by 18% and improving computational efficiency by 3× over baseline heuristics for TSP-class problems.	

Projects

E-Commerce Data Intelligence Pipeline <i>MfilterIt Software Development Engineer</i>	June 2025 – Present
- Built distributed scrapers on AWS Lambda and EC2 processing 20M+ URLs in under 4 hours across Blinkit, Zepto, Flipkart, Amazon, and Noon; reverse-engineered mobile app APIs using Frida, Charles Proxy, and HTTP Toolkit. - Designed a Kafka streaming layer batching 10K IDs/cycle with fault-tolerant upsert logic; built a multi-stage PySpark ETL pipeline processing 20M+ daily records. - Merged ecommerce and campaign datasets to compute KPIs; exposed analytics via a FastAPI dashboard with role-based access control.	
Weapon-Target Assignment Optimization <i>DayalBagh Educational Institute, Agra</i>	October 2024 – May 2025
Developed a metaheuristic optimization framework for the Static Weapon-Target Assignment Problem (SWTAP) using a Biased Random-Key Genetic Algorithm (BRKGA) with tailored crossover and mutation operators, achieving computationally efficient constraint satisfaction in C/C++.	

Publications

- “The Mobile Healthcare Vehicle Traveling Salesman Problem with Constant Budget Constraint: Heuristics and Hybrid Evolutionary Algorithms” — under review, *Journal of the Institution of Engineers (India) Series B*, Springer.

Education

DayalBagh Educational Institute, Agra <i>B.Tech in Electrical Engineering, Specialization: Computer Science</i>	2021–2025 8.8 CGPA (Top 10%)
REI Intermediate College, Agra <i>Senior Secondary (PCM)</i>	2020–2021 97.2%
REI Intermediate College, Agra <i>High School</i>	2018–2019 95.5%